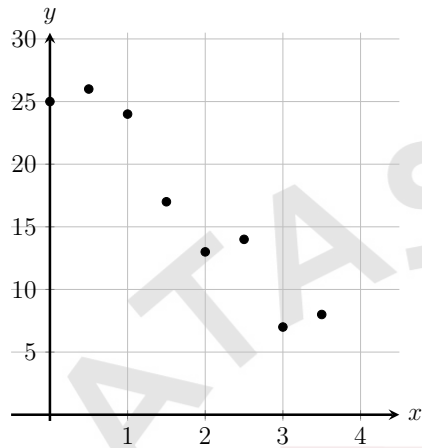


Scatterplots

1. [June 2023]

The scatterplot shows the relationship between two variables, x and y .

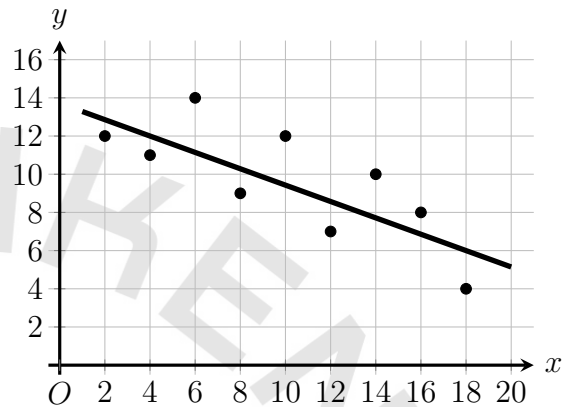


Which of the following equations is the most appropriate linear model for the data shown?

- A) $y = -6 + 28x$
- B) $y = 6 - 28x$
- C) $y = 28 + 6x$
- D) $y = 28 - 6x$

2. [August US 2023]

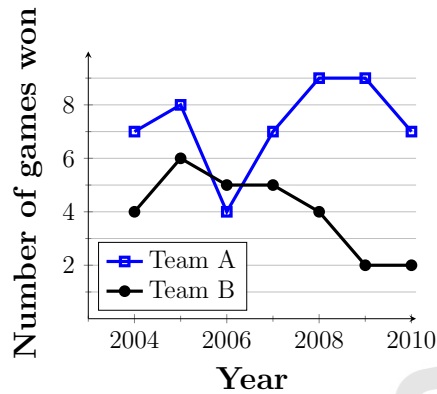
The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



What data point has an actual y -value that is 2 less than the y -value predicted by the line of best fit for corresponding x -value?

- | | |
|-----------|------------|
| A) (2,12) | C) (10,12) |
| B) (6,14) | D) (18,4) |

3. [August US 2023]



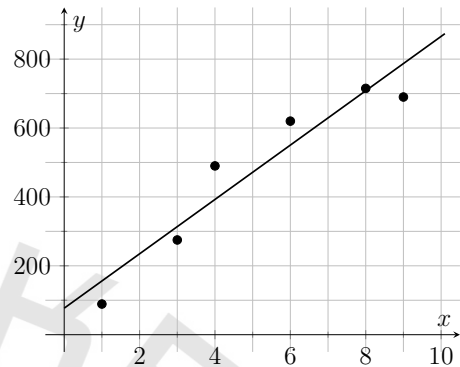
The line graphs show the number of games won by two football teams from 2004 to 2010. For how many of the years shown was the number of games won by Team *A* greater than the number of games won by Team *B*?

- A) 1
- B) 2

- C) 6
- D) 7

4. [March 2024]

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.

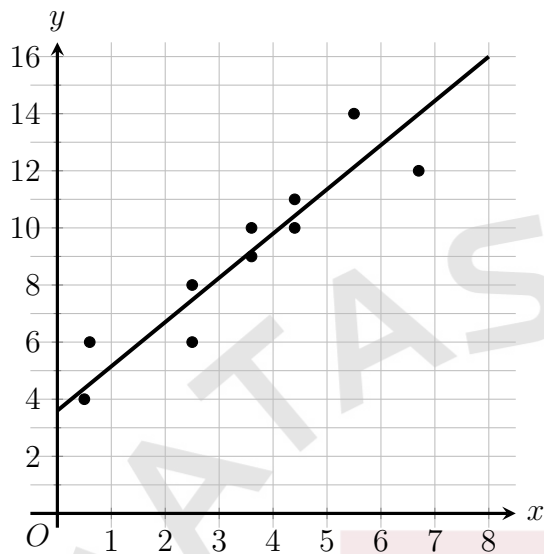


Which of the following best represents the line of best fit shown?

- A) $y = 77 + 0.01x$
- B) $y = 77 + 0.79x$
- C) $y = 77 + 68.9x$
- D) $y = 77 + 78.9x$

5. [March 2024]

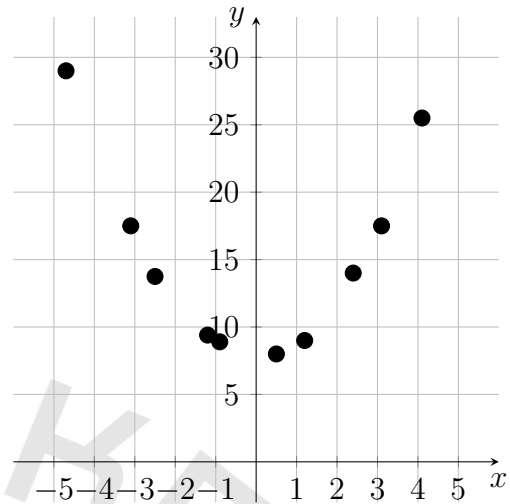
The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

- A) $y = 3.5 + 1.8x$
- B) $y = -2.4 + 1.55x$
- C) $y = 3.4 + 1.55x$
- D) $y = 2.4 - 1.5x$

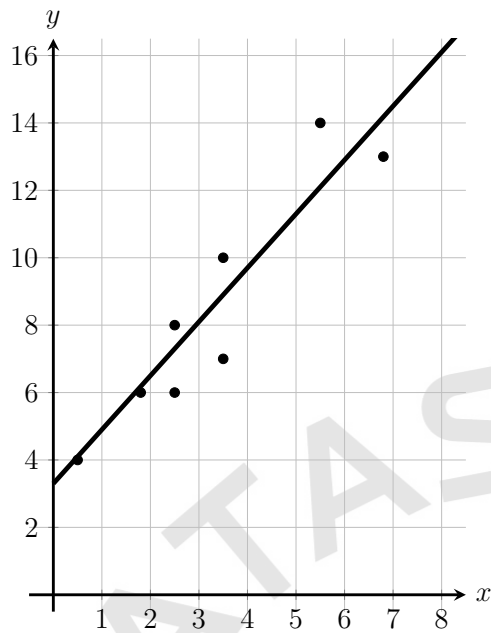
6. [March 2024]



The scatterplot shows the relationship between two variables, x and y . Which of the equations is the most appropriate model for the data shown?

- A) $y = x^2 - 8$
- B) $y = x^2 + 8$
- C) $y = (x - 8)^2$
- D) $y = (x + 8)^2$

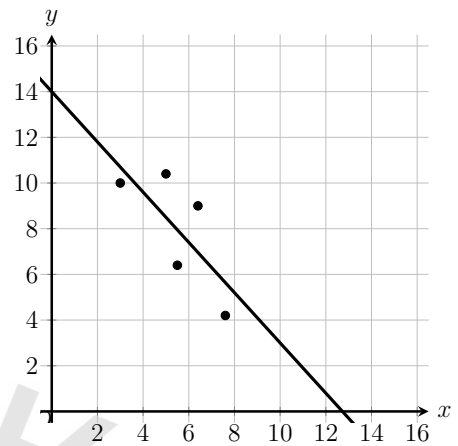
7. [May US 2024]



Which of the following equations best represents the line of best fit shown?

- A) $y = 3.3 + 1.6x$
- B) $y = 3.3 - 1.6x$
- C) $y = -3.3 + 1.6x$
- D) $y = -3.3 - 1.6x$

8. [October 2024]

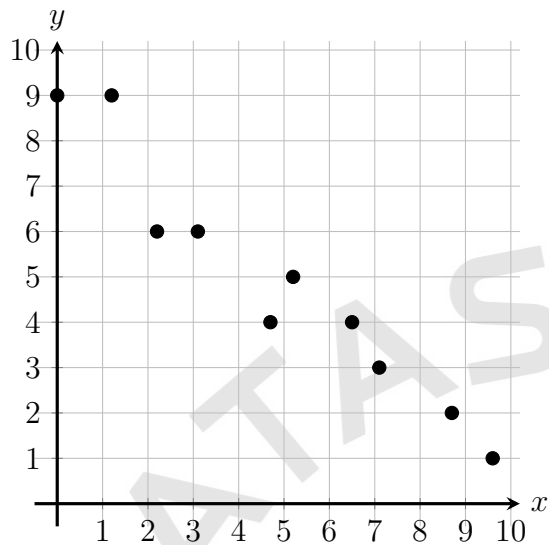


Which of the following is closest to the slope of this line of best fit?

- | | |
|-----------|----------|
| A) -3.3 | C) 1.1 |
| B) -1.1 | D) 3.3 |

9. [June 2024]

The scatterplot shows the relationship between two variables, x and y .

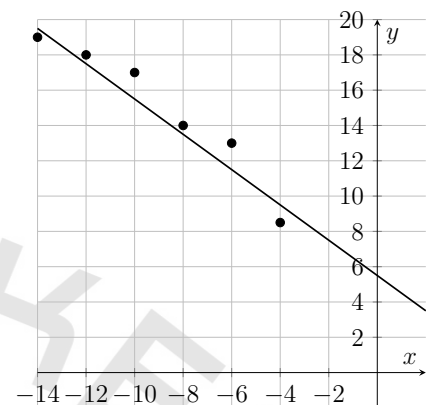


Which of the following equations is the most appropriate linear model for the data shown?

- A) $y = 0.8 + 8.7x$
- B) $y = 0.8 - 8.7x$
- C) $y = 8.7 + 0.8x$
- D) $y = 8.7 - 0.8x$

10. [August 2024]

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.

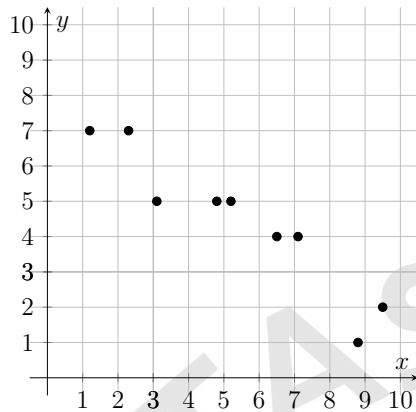


Which of the following equations best represents the line of best fit shown?

- A) $y = x - 19.9$
- B) $y = x - 19.9$
- C) $y = x - 5.7$
- D) $y = -x + 5.7$

11. [June 2024]

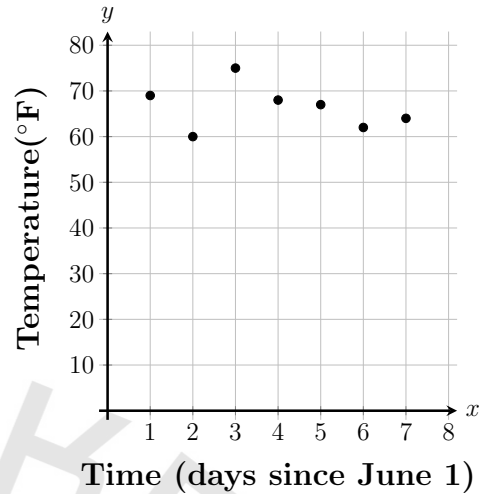
The scatterplot shows the relationship between two variables, x and y .



Which of the following equations is the most appropriate linear model for the data shown?

- A) $y = 0.8 + 8.7x$
- B) $y = 0.8 - 8.7x$
- C) $y = 8.7 + 0.8x$
- D) $y = 8.7 - 0.8x$

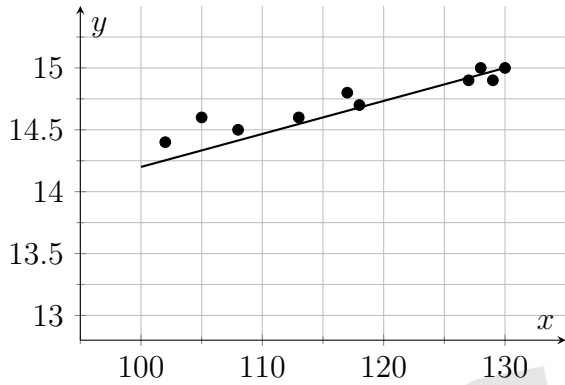
12. [August 2024]



During which of the following time periods did the greatest increase in recorded temperature take place?

- A) From $x = 6$ to $x = 7$
- B) From $x = 5$ to $x = 6$
- C) From $x = 2$ to $x = 3$
- D) From $x = 1$ to $x = 2$

13. [August US 2024]

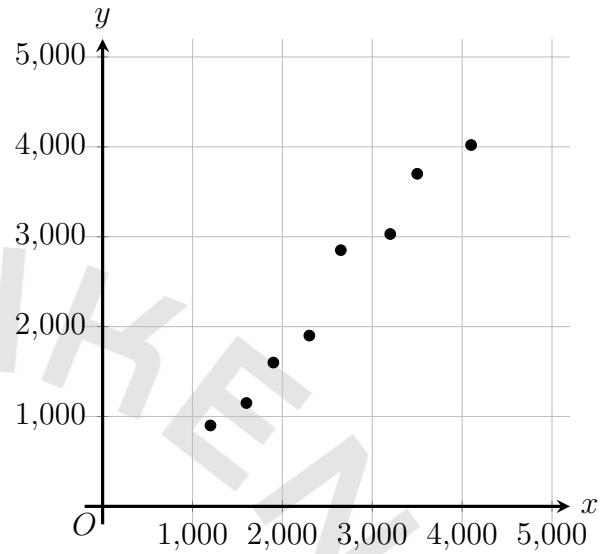


The scatterplot shows the relationship between two variables, x and y , for the dataset P . A line of best fit for the data is also shown. Data set Q is created by adding 540-units to the value of x for each data point from the data set P . Which of the following could be an equation of a line of best fit for the data set Q ?

- A) $-3 + \frac{35}{1301}x$
- B) $9 + \frac{1}{131}x$
- C) $18 + \frac{1}{131}x$
- D) $26 + \frac{35}{1301}x$

14. [October US 2024]

The scatterplot shows the relationship between two variables, x and y .

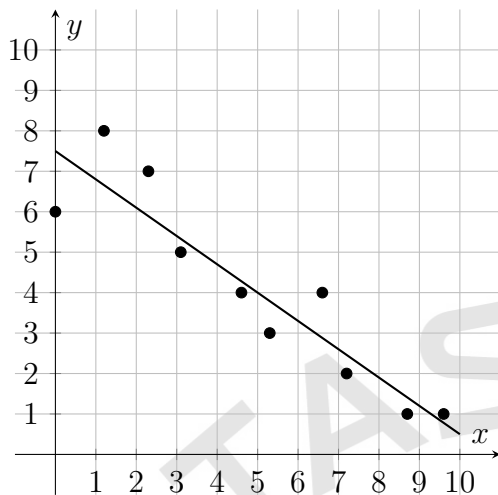


Which of the following equations is the most appropriate model for the data shown?

- A) $y = -582 + 1.2x$
- B) $y = 582 - 1.2x$
- C) $y = -582(1.2)^x$
- D) $y = 582(1.2)^x$

15. [October US 2024]

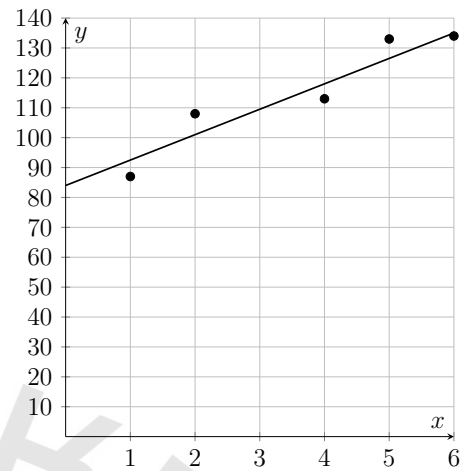
In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of this line of best fit?

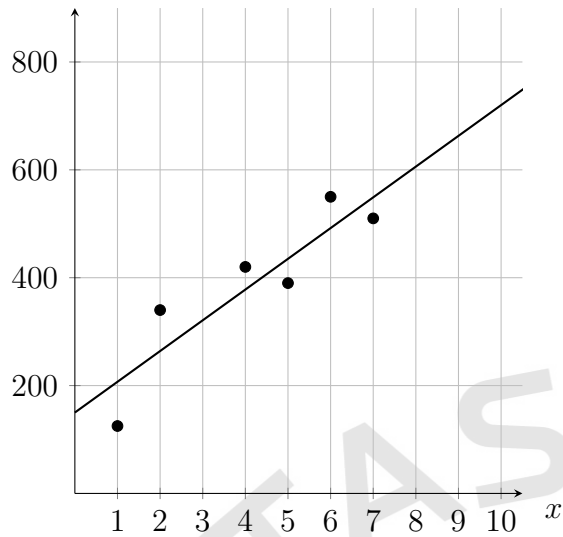
- A) -7 C) 0.7
 B) -0.7 D) 7

16. [October US 2024]



The scatterplot shows the body length, in centimeters (cm), of a New Zealand fur seal from an age of 1 to 6 years old. A line of best fit is also shown. For a New Zealand fur seal at an age of 3 years old, what is the body length in centimeters predicted by the line of best fit, to the nearest 10 cm ?

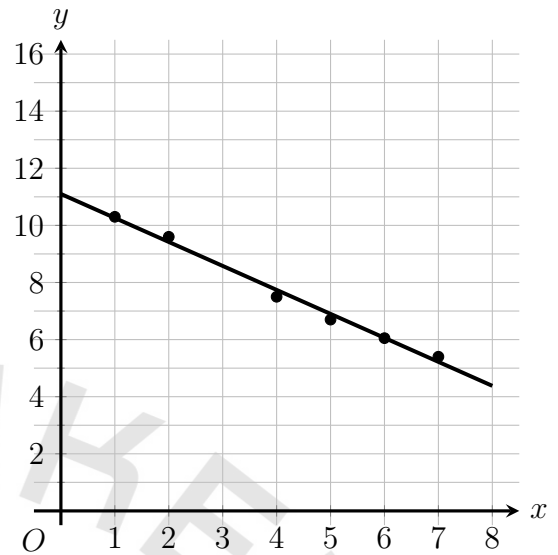
17. [November 2024]



Which of the following best represents the line of best fit shown?

- A) $y = 148.13 + 0.02x$
- B) $y = 148.13 + 0.59x$
- C) $y = 148.13 + 48.61x$
- D) $y = 148.13 + 58.61x$

18. [December 2024]

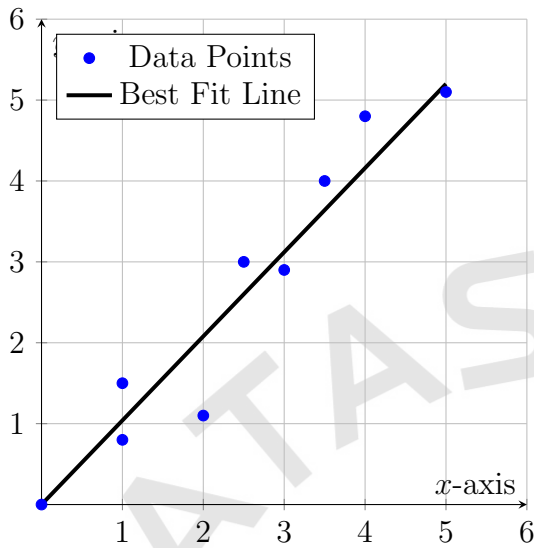


Which of the following is closest to the slope of the line of best fit shown?

- | | |
|-------------|------------|
| A) -11.19 | C) 0.84 |
| B) -0.84 | D) 11.19 |

19. [December 2024]

In the given scatterplot, a line of best fit for the data is shown.

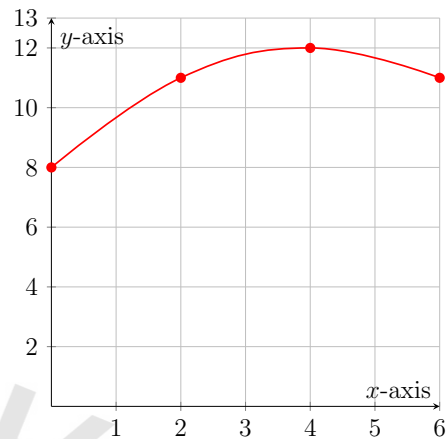


Which of the following equations best represents this line of best fit?

- A) $y = 1.1x$
 B) $y = -1.1x$

- C) $y = 5.2$
 D) $y = -5.2$

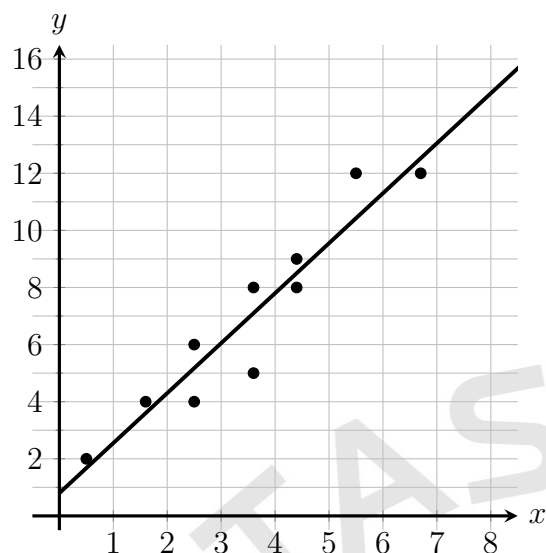
20. [December 2024]



The graph models the number of active projects a company was working on x months after the end of November 2011, where $0 \leq x \leq 6$. According to the model, what is the predicted number of active projects the company was working on at the end of November 2011?

- A) 0
 B) 8
 C) 11
 D) 12

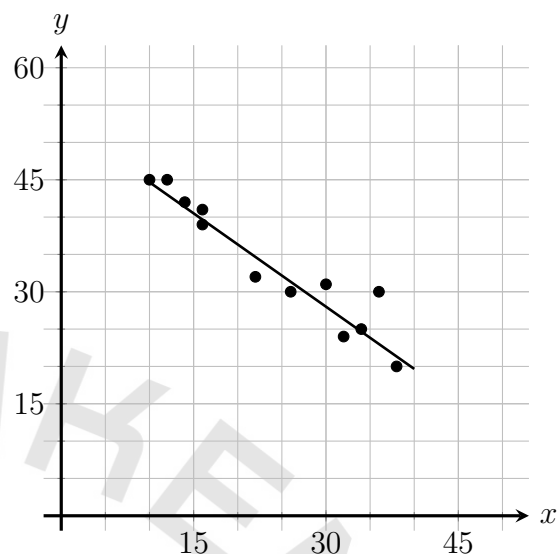
21. [March 2025]



Which of the following equations best represents the line of best fit shown?

- A) $y = 0.8 + 1.7x$
- B) $y = 0.8 - 1.7x$
- C) $y = -0.8 + 1.7x$
- D) $y = -0.8 - 1.7x$

22. [March US 2025]

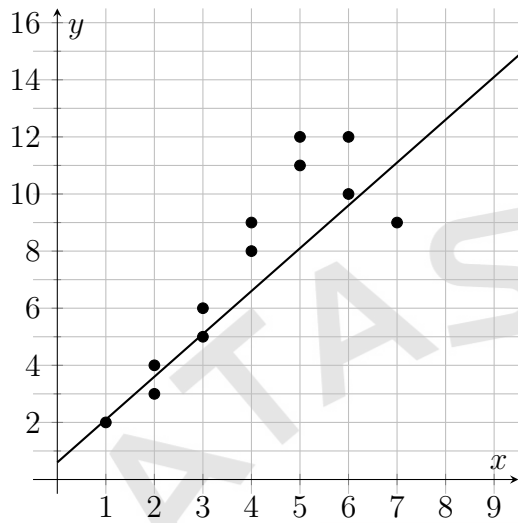


The scatterplot shows the relationship between two variables, x and y , for data set A . A line of best fit for the data is also shown. Data set B is created by subtracting 9 units from the value of y for each data point from data set A . Which of the following is closest to the y -coordinate of the y -intercept of the **line** of best fit for data set B ?

- | | |
|----------|----------|
| A) 45.57 | C) 59.62 |
| B) 53 | D) 62.75 |

23. [March 2025]

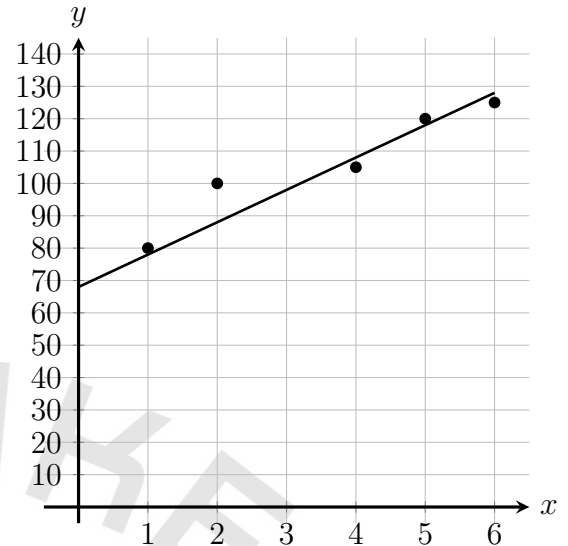
The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

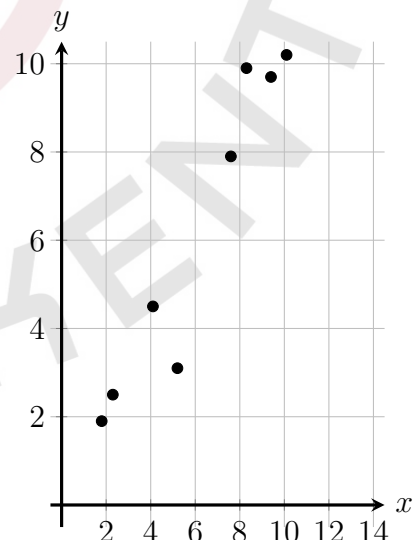
- A) $y = 0.6 + 1.5x$
- B) $y = 0.6 - 1.5x$
- C) $y = -0.6 + 1.5x$
- D) $y = -0.6 - 1.5x$

24. [June US 2025]



The scatterplot shows 5 measurements of the body length, in centimeters (cm), of a New Zealand fur seal from an age of 1 year to 6 years old. A line of best fit is also shown. For a New Zealand fur seal at an age of 3 years old, what is the body length predicted by the line of best fit, to the nearest 10 cm ?

25. [November 2023]



Which of the following equations is the most appropriate linear model for the data shown?

- A) $y = -2x$
- B) $y = -x$
- C) $y = x$
- D) $y = 2x$

Questions 26 and 27 refer to the following information

The line graph shows the amount, in millions of dollars, that the National Science Foundation (NFS) granted to museums in Minnesota for the years 2000 through 2006.



26. [August US 2023]

In 2002, the NFS granted a total of approximately 39 million dollars to US museums. Which of the following is closest to the percentage of this total that was granted to museums in Minnesota?

- A) 0.1%
- B) 1.2%
- C) 8.2%
- D) 12.2%

27. [August US 2023]

What is the approximate average rate of change of the amount, in millions of dollars per year, that the NFS granted to museums in Minnesota from 2000 to 2006?

- A) -2.07
- B) -0.48
- C) 0.41
- D) 0.82